

## INSTRUMENTAL VARIABLES

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Properties of  $\hat{\beta}_{LSE}$  under different scenarios:

- ① Non-stochastic regressors, assumptions of the CLRM hold, then
  - $\hat{\beta}_{LSE}$  is unbiased, consistent. Further, if  $\epsilon \sim N(0, \sigma^2 I)$ , then  $\hat{\beta}_{OLS} \sim N(\beta, \sigma^2 (X'X)^{-1})$  (and similarly for  $\hat{\beta}_{NLS}$ ,  $\hat{\beta}_{GLS}$ , etc.), exact  $t$ ,  $F$  tests in diff. testing problems
- ② Stochastic regressors but uncorrelated with error (weak exogeneity property), then
  - $\hat{\beta}_{LSE}$  unbiased, consistent, etc.
- ③ Stochastic regressors, asymptotically uncorrelated with error, but correlated in small/finite sample size — more realistic situation, then

For  $y = X\beta + \epsilon$ ,  $E(\epsilon) = 0$ ,  $Cov(\epsilon) = \sigma^2 I$ , say, so that  $\hat{\beta}_{OLS}$  is unbiased.

$$\begin{aligned}\hat{\beta}_{OLS} &= (X'X)^{-1} X'y = (X'X)^{-1} X'(X\beta + \epsilon) \\ &= \beta + \left(\frac{X'X}{T}\right)^{-1} \left(\frac{X'\epsilon}{T}\right)\end{aligned}$$

With the regularity conditions, viz, (i)  $\lim_{T \rightarrow \infty} \frac{X'\epsilon}{T} = 0$  and (ii)  $\lim_{T \rightarrow \infty} \frac{X'X}{T} = Q$ , p.d.

it is possible to have  $\lim_{T \rightarrow \infty} \hat{\beta}_{OLS} = \beta + \lim_{T \rightarrow \infty} \left(\frac{X'X}{T}\right)^{-1} \lim_{T \rightarrow \infty} \left(\frac{X'\epsilon}{T}\right) = \beta + Q^{-1} \cdot 0 = \beta$ , i.e.,  $\hat{\beta}_{OLS}$  consistent.

here  $\hat{\beta}_{OLS} \overset{asy}{\sim} N\left(\beta, \sigma^2 \frac{Q^{-1}}{T}\right)$ ,  $\frac{1}{\sqrt{T}} \hat{\beta} \overset{asy}{\sim} N\left(\frac{1}{\sqrt{T}} \beta, \frac{\sigma^2}{T} Q^{-1}\right)$ , etc.

- ④ Stochastic regressors & correlated with error even asymptotically, i.e., regularity condition  $\lim_{T \rightarrow \infty} \frac{X'\epsilon}{T} = 0$  fails, and  $\lim_{T \rightarrow \infty} \hat{\beta} \neq \beta$ ,  $\hat{\beta}_{LSE}$  is inconsistent (not unbiased even asymptotically).

Then we consider, the Method of Instrumental Variables (IV) Estimation

## Example of inconsistent $\hat{\beta}$ in OLS.

### Permanent Income Hypothesis

$y_t = \alpha + \beta x_t^* + \epsilon_t$ , where  $y_t$  = current consumption exp.  
 $x_t^*$  = permanent income, which is diff. to observe  
 so, we observe  $x = x^* + \omega$ ,  $\omega$  = error in meas. of  $x^*$

Thus, we estimate,  $y_t = \alpha + \beta(x_t - \omega_t) + \epsilon_t$   
 $= \alpha + \beta x_t + (\epsilon_t - \beta \omega_t)$   
 $= \alpha + \beta x_t + \tilde{\epsilon}_t$ , even if  $\epsilon_t$  are classical errors.

$$Cov(x_t, \tilde{\epsilon}_t) = Cov(x_t, \epsilon_t - \beta \omega_t) = 0 - \beta Cov(x_t^* + \omega, \omega) = -\beta \sigma_{\omega}^2 \neq 0.$$

(Inconsistency brought about by use of proxy variable & measurement error).

Consider the UR model,  $y_t = \beta' z_t + \epsilon_t$ ,  $t=1(1)T$   
 let  $\tilde{z}_t^{k \times 1}$  = vector of instruments/instrumental variables at fixed pt.  $t$   
 (one instrument for each regressor)

$\tilde{z}_t$  must satisfy:

- (i) Instruments should be uncorrelated with error term, i.e.,  
 $E(\epsilon_t \tilde{z}_t) = 0$  if  $t=1(1)T$
- (ii) Instruments should be highly correlated with the regressors.

The 2 standard regularity condns, viz., (i)  $\lim_{T \rightarrow \infty} \frac{X'X}{T} = Q$ , p.d. and  
 (ii)  $\lim_{T \rightarrow \infty} \frac{X'\epsilon}{T} = 0$  to be

replaced by 3 similar set of restrictions

(i)  $\lim_{T \rightarrow \infty} \frac{\tilde{Z}'X}{T} = Q_{ZX}$ , finite, n.s.

(ii)  $\lim_{T \rightarrow \infty} \frac{\tilde{Z}'\tilde{Z}}{T} = Q_{ZZ}$ , p.d.

(iii)  $\lim_{T \rightarrow \infty} \frac{\tilde{Z}'\epsilon}{T} = 0$

When  $\text{plim} \frac{X'E}{T} \rightarrow 0$ ,  $\hat{\beta}_{OLS}$  not consistent, then apply IV estimation using the 3 regularity cond.

Method 1  $y = X\beta + \epsilon$ , with  $E(X'E) \neq 0$

Define the IV estimator  $\hat{\beta}_{IV} = (Z'X)^{-1} Z'y$   
 $= (Z'X)^{-1} Z'(X\beta + \epsilon)$   
 $= \beta + (Z'X)^{-1} Z'\epsilon$

so that  $\text{plim}_{T \rightarrow \infty} \hat{\beta}_{IV} = \beta + \text{plim}_{T \rightarrow \infty} \left( \frac{Z'X}{T} \right)^{-1} \text{plim}_{T \rightarrow \infty} \left( \frac{Z'\epsilon}{T} \right) = \beta + TQ_{ZX}^{-1} 0 = \beta$   
 $\Rightarrow \hat{\beta}_{IV}$  consistent for  $\beta$

Method 2 Two-stage least sq. (2SLS)

$$y = X^{TXK} \beta_{K \times 1} + \epsilon^{TX1}$$

$X^{TXK}$ : matrix of  $k$  regressors, further  $Z^{TXk}$ : matrix of  $k$  instruments

Stage 1 Obtain prediction of  $X$  vals from regressing them on  $Z$  vars.  
 Multivariate MLR model set-up  $X$  dep/endogenous,  $Z$  purely exogenous

$$X = Z\Pi + \epsilon_{TXK}, \quad \Pi_{K \times K}: \text{matrix of parameters}$$

$\epsilon_{TXK}$ : matrix of errors.

(Reduced form/auxiliary eqs for endogenous vars)

$$\hat{\Pi}_{OLS} = (Z'Z)^{-1} Z'X, \quad \text{and } \hat{X} = Z\hat{\Pi} = Z(Z'Z)^{-1} Z'X = P_Z X$$

$P_Z$  projection matrix on  $\mathcal{C}(X)$ ,  $P_Z$  symmetric & idempotent

$X$  is even asymptotically correlated with error  $\epsilon$ ,  
 but  $\hat{X}$  asymptotically uncorrelated with error (prove this!)

So if we use  $\hat{X}$  as regressors (in place of  $X$ ), then  $\hat{\beta}_{OLS}$  will be consistent.

Stage 2  $y = \hat{X}\beta + \epsilon$ ,  $\hat{\beta}_{OLS}$  of  $y$  on  $\hat{X}$   
 $= \hat{\beta}_{2SLS} = (\hat{X}'\hat{X})^{-1} \hat{X}'y$   
 $= \hat{\beta}_{IV}$

Method 3

Perform stage 1 of 2SLS,

$$Y = X\beta + \epsilon, \quad X = Z\pi + \epsilon.$$

Obtain  $\hat{\pi}_{OLS}$  from  $X = Z\pi + \epsilon$ ,

Obtain  $\hat{X}$ . Use  $\hat{X}$  as instruments in place of  $Z$  in 2SLS estimator.

$$\text{Then } \hat{\beta}_{IV} = (\hat{X}'\hat{X})^{-1}\hat{X}'Y = (Z'X)^{-1}Z'Y \quad \text{use } \hat{X} = P_Z X.$$

Method 4 Residual Augmented Regression,

$$Y = X\beta + \epsilon \\ X = Z\pi + \epsilon.$$

Stage 1 Perform stage 1 of 2SLS

$$\text{Obtain } \hat{\pi}_{OLS}, \hat{X} \text{ and } \hat{\epsilon} = (X - Z\hat{\pi}) = (I - P_Z)X$$

$\hat{X}, \hat{\epsilon}$  are uncorrelated

Stage 2 Regress  $Y$  on  $\hat{X}$  &  $\hat{\epsilon}$

$$Y = \hat{X}\beta + \hat{\epsilon}\delta + u \quad \text{to get,}$$

$$\begin{pmatrix} \hat{\beta}_{IV} \\ \hat{\delta} \end{pmatrix} = \begin{bmatrix} (\hat{X}'\hat{X}) & \hat{X}'\hat{\epsilon} \\ \hat{\epsilon}'\hat{X} & \hat{\epsilon}'\hat{\epsilon} \end{bmatrix}^{-1} \begin{pmatrix} \hat{X}'Y \\ \hat{\epsilon}'Y \end{pmatrix} = \begin{pmatrix} \hat{X}'\hat{X} & 0 \\ 0 & \hat{\epsilon}'\hat{\epsilon} \end{pmatrix}^{-1} \begin{pmatrix} \hat{X}'Y \\ \hat{\epsilon}'Y \end{pmatrix}$$

$$= \begin{pmatrix} (\hat{X}'\hat{X})^{-1}\hat{X}'Y \\ (\hat{\epsilon}'\hat{\epsilon})^{-1}\hat{\epsilon}'Y \end{pmatrix}.$$

$$\Rightarrow \hat{\beta}_{IV} = (\hat{X}'\hat{X})^{-1}\hat{X}'Y = [(P_Z X)'(P_Z X)]^{-1}(P_Z X)'Y$$

$$= (X'P_Z X)^{-1}X'P_Z Y$$

$$= (Z'X)^{-1}Z'Y$$

Just Identified case as many instruments as regressors

Over Identified case more instruments than regressors.

For one explanatory variable  $\exists$  2 instruments which are equally good.

$X_{T \times k}$  : regressors ;  $Z_{T \times p}$  : instruments  $p > k$ .

Regress  $X$  on  $Z$ ,  $X = Z\pi + \epsilon$  ; Obtain  $\hat{X} = P_Z X$

$$\text{Then } \hat{\beta}_{IV} = (\hat{X}'\hat{X})^{-1}\hat{X}'Y =$$

Use  $\hat{X}$  as instruments

## Practical issues

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### ① Choice of instruments

Out of  $k$  regressors  $x_1, \dots, x_k$ , only

$l$  are endogenous

$$x_{T \times l}^{(2)} = (x_{k-l+1}, \dots, x_k)$$

& rest  $k-l$  are exogenous

$$x_{T \times k-l}^{(1)} = (x_1, \dots, x_{k-l})$$

$$\text{Wlog. } \Rightarrow X = (x^{(1)} \ x^{(2)})$$

Note

i)  $x^{(1)}$  will contain the info on intercept, dummies, time trends etc.  
(If a variable is exogenous, it is its own instrument - satisfies all 3 reg. condns)

ii) total # of instruments  $\geq k$ , so choose instruments only for the endogenous, & repeat the exogenous in  $Z$ .

### Ex Choice of instruments for regr. in $x^{(2)}$

Demand for perishable goods (like, agricultural goods, dairy products, etc.)  
 $\ln Q_{d,t} = \beta_0 + \beta_1 \ln P_{t-1} + \epsilon_t$  ( $\because$  here demand at  $t-1$  determines supply in  $t$ ,  
 $\therefore \epsilon_t$  contains  $Q_{s,t-1}$ , which is correlated with  $P_{t-1}$ )  
using market equilibrium,  $Q_d = Q_s$ )

Data = price & qty (supplied) of goods

$\beta_1$  = price elasticity of qty demanded.

Really, the model should include var's like price of substitutes, complements, related goods (law of demand!).

Can these serve as instruments?

Let  $R_t$  = rain fall at time  $t$ . Is it a valid instrument?

- i)  $\text{Cor}(R_t, \epsilon_t) = 0$  (whether it rains or not, should not affect demand)
- ii)  $\text{Cov}(R_t, P_t) \neq 0$  (flood or drought affects prices of perishable products instantaneously)

then  $x^{(1)} = (1)$   $x^{(2)} = (P)$ ,  $Z = (1 \ R)$ .

~~But before this~~  
② Whether IV method is reqd, is OLS not good enough?  
If yes then whether chosen instruments are valid?